



Multilingual Health Question Answering in Low-Resource African Languages

Course : Machine Learning Techniques I

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Abstract

This report documents the experimentation and development of a multilingual sequence-to-sequence model for answering health related questions in low resource African languages as part of Zindi Multilingual Health Question Answering Challenge by ITU. The task requires generating accurate, fluent and contextually appropriate health in eight language subsets including four genuinely low-resource African languages like Akan(Ghana), Luganda(Uganda), Swahili(Kenya), and Amharic(Ethiopia). This work finetune google/mt5-small using Hugging Face Transformers library and documents over 10 systematic experiments including preprocessing strategies, generation parameter optimisation, hyperparameter tuning and model configuration. The key findings include the critical instability of fp 16 mixed precision training with mT5 on T4 GPUs, the effectiveness of language tag prefixing for multilingual generation and the unique challenges posed by Amharic due to its distinct Ge'ez script and limited training data. Evaluation uses a weighted combination of ROUGE-1 F1 (0.37), ROUGE-L F(0.37) and LLM as a judge(0.26).

1. Project Overview

In sub-Saharan Africa access to reliable health information remains a critical challenge. Language barriers frequently stop communities from receiving accurate and clear guidance on health in their native languages, particularly on sensitive topics such as sexual, maternal and reproductive health(MSRH). This Zindi competition organized by the international

Telecommunication Union (ITU) challenges participants to build a multilingual model capable of accurately answering health related questions in low resource African languages.

The task is mounted as a conditional text generation problem : given a health question posed in any of eight supported language/country subsets, the model must generate an accurate, fluent and contextually appropriate response in the same language. The dataset covers Akan(Ghana), Luganda(Uganda), Swahili(Kenya), Amharic(Ethiopia) and multiple English variants across Uganda, Ghana,Ethiopia and Kenya.

This project uses google/mt5-small as a multilingual encoder-decoder backbone, fine tuned end to end using the HuggingFace Seq2Seq Trainer. A language tag prefix strategy is applied to all inputs to condition the model on the target output language and locale. The full experimentation pipeline is documented across 10+ meaningful experiments covering preprocessing, training configuration, hyperparameter tuning and generation strategy.

2 Dataset Understanding and Exploratory Data Analysis

2.1 Dataset Structure

The dataset consists of three files : Train.csv with **29,815 rows**, Test.csv with **2,618 rows** and SampleSubmisiion.csv. Each row contains four columns : ID(unique identifier), input(the health question), output(the reference answer, available only in Train.csv) and subset(the language/country code identifying the language and country of origin).

2.2 Language Distribution

The dataset covers eight language/country subsets. Table1 presents the distribution across training and test sets along with average answer lengths derived from exploratory analysis.

Table 1: Dataset distribution by language subset

Subset	Language	Train Rows	Test Rows	Avg Answer Length (words)
Eng_Uga	English (Uganda)	7,624	744	95.4
Aka_Gha	Akan (Ghana)	4,455	492	105.6
Eng_Gha	English (Ghana)	4,443	491	75.1
Eng_Eth	English (Ethiopia)	3,915	60	24.5
Lug_Uga	Luganda (Uganda)	3,383	374	79.7
Eng_Ken	English (Kenya)	2,080	167	78.7
Swa_Ken	Swahili (Kenya)	2,070	229	84.3
Amh_Eth	Amharic (Ethiopia)	1,845	61	20.2
Total		29,815	2,618	—

2.3 Key EDA Findings

Several important insights emerged from exploratory analysis :

Language imbalance : 60% of training rows (8,062) approximately are English across four countries while 40% only are low resource African languages. This imbalance means the model wil naturally perform better on english subsets without explicit mitigation strategies.

Answer length variation : Akan answers average 105.6 words per response, substantially longer than Amharic(20.2 words) and english-ethiopia(24.5 words). Since the default max_output_len of 128 tokens may truncate longer Akan answers, this became the basis of experiment 5 (increased max_output_len).

Amharic as the hardest case : amharic has the smallest training set of 1,845 examples, the fewest test examples of 61 and uniquely uses the Ge'ez script rather than Latin characters. This combination makes it the highest risk language for poor performance and the most important to flag in limitations.

Data quality issues : manual inspection of validation predictions reveals at least one mislabeled example where the reference answer for a question about syphilis referenced herpes instead. This suggests some noise in the training labels which may affect evaluation scores independently of model quality.

Test distribution mirrors training : the test language distribution roughly proportional to training suggesting no significant domain shift between train and test sets

3 Fine-Tuning Methodology

3.1 Model selection

The base model for this task is Google mT5-small. mT5 is a massively multilingual variant of T5 pre-trained on the mC4 corpus covering 101 languages, making it a natural fit for a task spanning African and English languages. The small variant (300M parameters) was chosen for practical GPU constraints fine-tuning on a free-tier Kaggle T4 GPU (15GB VRAM) with full fp32 precision. Larger variants (mT5-base, mT5-large) were considered for later experiments to assess the impact of model scale.

3.2 Preprocessing Pipeline

Language-tag prefix: each input question is prepended with a language/locale identifier using the format "answer in {subset}: {question}" for example, "answer in Aka_Gha: Okwan bən so na mmabun..." This strategy conditions the encoder on both the content of the question and the target language, enabling a single model to generate responses in the correct language without separate language-specific models. This is a standard technique for multilingual seq2seq tasks and was adopted as the core preprocessing decision.

Train/validation split: the 29,815 training examples were split 90/10 (26,833 train, 2,982 validation) using stratified sampling on the subset column, ensuring all eight languages are proportionally represented in both splits.

Tokenization :P inputs and outs were tokenized using mT5 SentencePiece tokenizer with **max_length of 128** for both input and output sequences. Padding tokens in labels were replaced with -100 so the cross-entropy loss function ignores them during training. Max sequence **lengths were initially set to 128** as a baseline with experiment 5 exploding the effect of **increasing max_output_leng of 256 to accommodate longer Akan answers.**

3.3 Training configuration

All experiments use HuggingFace's Seq2SeqTrainer with the following baseline configuration unless otherwise noted: learning rate 3e-4, batch size 4 per device (2 T4 GPUs), fp32 precision (fp16 disabled after Experiment 1 failure), AdamW optimizer (default), and predict_with_generate=True for ROUGE-compatible evaluation. The output directory is

set to /kaggle/working/ to avoid disk quota issues encountered during early checkpointing experiments.

3.4 Evaluation Metrics

The competition evaluates submissions using a weighted combination of three metrics: ROUGE-1 F1 (weight 0.37), measuring unigram overlap between predicted and reference answers; ROUGE-L F1 (weight 0.37), measuring longest common subsequence similarity; and LLM-as-a-Judge (weight 0.26), where a large language model scores predictions on factual accuracy, completeness, and language appropriateness on a 1–5 scale normalized to [0,1]. Internal validation during training tracks cross-entropy loss rather than ROUGE, since computing ROUGE at every evaluation step is computationally expensive with a large dataset.

4. Experiment Progression and Tracking

A total of 10 meaningful experiments were conducted and documented. Table 2 provides a summary; detailed analysis of each follows.

Table 2: Experiment Summary

Exp #	Change Made	Train Loss	Val Loss / LB Score	Key Insight
1	mT5-small, fp16=True, 3 epochs, lr=3e-4	0.0000 (NaN)	NaN	FAILED: fp16 causes NaN with mT5 on T4 — use fp32
2	fp16=False (fp32), 2 epochs, batch=8	6.06	Val: 4.28	Training stable; model generates plausible but shallow answers
3	fp32, 4 epochs, batch=4	3.98	Val: 3.48 / LB: 0.255044	Best result — longer training improves multilingual output quality
4	num_beams=1 (greedy) vs num_beams=4	—	Planned	Not completed due to GPU quota and time constraints
5	max_output_len=256 (Akan truncation fix)	—	Planned	Not completed due to GPU quota and time constraints
6	No language prefix (ablation)	—	Planned	Not completed due to GPU quota and time constraints

7	Lower learning rate: 1e-4 vs 3e-4	—	Planned	Not completed due to GPU quota and time constraints
8	Per-language ROUGE evaluation	—	N/A (analysis only)	Planned analysis to reveal per-language performance gaps
9	length_penalty=2.0 in generation	—	Planned	Not completed due to GPU quota and time constraints
10	save_strategy=no (disk optimization)	—	N/A (infra fix)	Implemented in Exp 3 — prevents disk-full crash during training

Experiment 1 : fp16fine-tuning(failed)

Configuration : mT5-small, fp16=true, 3 epochs, lr=3e-4, batch size 8, language prefix inputs.
Outcome: training loss collapsed to exactly 0.0000 across all three epochs with validation loss that reports to NaN. the TrainOutput summary confirmed training loss = 0.0

Insight: fp16 mixed-precision training is a known source of numerical instability with mT5 models on consumer-grade GPUs (T4). The mT5 architecture has tied embedding weights that can produce NaN gradients under fp16. This experiment establishes a critical baseline constraint: all subsequent experiments must use fp32 (fp16=False, bf16=False). This is a well-documented issue in the HuggingFace community and is noted here explicitly so that any reproduction of this work can avoid the same failure.

Experiment 2: fp32 Baseline, 2 Epochs

Configuration: mT5-small, fp16=False, 2 epochs, lr=3e-4, batch size 8 per device (dual T4), language-prefix inputs.

Outcome: training loss 6.07→4.28 (validation), decreasing smoothly across both epochs. Qualitative inspection shows the model generating text in the correct target language with plausible health related content but answers were often occasionally repetitive and short(eg :”HIV is a virus that attacks the immune system which attacks immune system”)

Insight: 2 epochs is insufficient for this model and dataset size. The model has learned basic language routing and topic recognition but lacks the training signal to generate detailed complete answers. This motivated experiment 3

Experiment 3: fp32, 4 Epochs (Best Result)

Configuration: mT5-small, fp16=False, 4 epochs, lr=3e-4, batch size 4 per device, save_strategy=no, language-prefix inputs.

Outcome: **validation loss decreased from 5.22 at step 1000 to 3.48 at step 13000 consistently lower than experiment 2.** The generated predictions showed correct language output with proper special characters(ε, ϐ in Akan; አጥፍኛ in Amharic) and more

complete health answers compared to experiment 2. Training took hours approximately on kaggle T4x2.

Insight : Doubling training epochs from 2 to 4 showed a meaningful improvement in validation loss (3.48 vs 4.28). The model benefits from more passes over the data to learn language specific generation patterns. However, loss is still relatively high suggesting that the model has more capacity to improve with additional epochs or larger model variants.

5 Results

5.1 Validation and training curves

Due to time constraints, the learning curve is presented as a table. Training loss decreased from 6.39 (step 1000) to 3.98 (step 13000). Validation loss decreased from 5.22 (step 1000) to 3.48 (step 13000) over 4 epochs."

Step vs Training Loss and Validation Loss for Experiment 3, showing smooth decrease from 6.39 to 3.98 (train) and 5.22 to 3.48 (val) across 13,420 steps / 4 epochs. This can be generated from the logged step/loss values in Table 2 using matplotlib.]

5.2 Leaderboard Score Progression

Table 3: Leaderboard Submission History

Submission #	Exp # Used	Public LB Score	Private LB Score	Rank	Notes
1	3	0.255044	0.257264	Under code review	First submission — 4-epoch mT5-small, language-prefix preprocessing

5.3 Qualitative Output Examples

Table 4 shows sample predictions from Experiment 3 across multiple languages to illustrate output quality.

Table 4: Sample predictions from Experiment 3

Subset	Input (truncated)	Predicted Answer (truncated)	Observation
Aka_Gha	Okwan ben so na mmabun...	Nneema a wode beye nkyerekyerε nneema, adwumay...	Correct Akan output with proper characters
Eng_Uga	How do needle exchange programs help reduce HIV?	No. needle exchange programs help reduce the risk of HIV transmission.	Correct language; answer too brief vs reference

Eng_Ken	How Long Does It Take for HIV to Cause AIDS?	HIV is a virus that attacks the immune system, which attacks the immune system.	Repetition issue; incomplete answer
Eng_Eth	How is syphilis diagnosed?	Syphilis is cured with antibiotics.	Correct topic but reference answer has labeling issue (says herpes)

6. Discussion and Critical Analysis

6.1 Why fp16 Failed with mT5

The most significant technical finding of this project is the complete failure of fp16 mixed-precision training with mT5-small on T4 GPUs (Experiment 1). The training loss collapsed to exactly 0.0 with NaN validation loss – a classic symptom of numerical overflow or underflow during the backward pass. This is a known issue with mT5's tied weight architecture where shared embedding matrices can produce NaN gradients when operating at half precision. Switching to fp32 fully resolved the issue at the cost of approximately 1.5x slower training and higher memory usage. This tradeoff is well justified given that a NaN loss produces a completely non-functional model.

6.2 Effect of Training Duration

Comparing Experiments 2 and 3 (2 epochs vs 4 epochs) shows a clear benefit to longer training: validation loss improved from 4.28 to 3.48 an improvement of approximately 19%. Qualitatively, the 4-epoch model generated more complete, grammatically consistent answers with correct special characters in Akan (ɛ, ɔ) and appropriate multilingual switching. This suggests the model was still learning at the end of 2 epochs and had not yet converged. However, validation loss was still decreasing at epoch 4, indicating further improvement might be achievable with additional training, subject to GPU hour constraints.

6.3 Language-Tag Prefix Strategy

Prepending "answer in {subset}: " to every input is a low-cost, high-value preprocessing decision motivated by established multilingual NLP practice. It serves two functions: it tells the decoder which language to generate in (language routing), and it provides additional context about the cultural/regional setting of the question. Experiment 6 (ablation without prefix) tests whether this prefix is actually necessary for mT5-small, which already has multilingual pretraining. The hypothesis is that the prefix significantly improves language consistency, particularly for languages underrepresented in mT5's pretraining corpus (Akan, Luganda).

6.4 Truncation and Akan Answer Length

Exploratory analysis revealed that Akan answers average 105.6 words substantially longer than the 128 token max_output_len used in baseline experiments. Since mT5's SentencePiece tokenizer produces approximately 1.2-1.5 tokens per word for Akan text (which has many special characters and compound words), a 128-token limit likely truncates many Akan answers mid-sentence. This motivated Experiment 5 (max_output_len=256), which is expected to improve ROUGE scores for Akan by generating more complete answers.

6.5 Infrastructure challenges

A significant portion of experimentation time was spent managing infrastructure constraints rather than model improvements. Key issues encountered as : (1) free tier colab GPU quota exhaustion after 3 hours approximately that requires migration to kaggle; (2) CUDA out of

memory errors when using DataParallel across 2 T4 GPUs with batch size 8 (resolved by reducing batch size to 4); (3) disk quota overflow when saving checkpoints every 500 steps (resolved by setting save strategy to no and saving only the final submission CSV); (4) session resets using trained model weights requiring full retraining. These challenges are documented here as they represent realistic engineering constraints in resource-constrained ML experimentation and their resolutions are incorporated into the reproducible notebook.

6.6 Limitations

Several important limitations of this work must be acknowledged:

- Model scale: mT5-small (300M parameters) is significantly smaller than state-of-the-art multilingual models. Larger models (mT5-base, mT5-large, or Aya-101) would likely produce substantially better answers but were not feasible within free-tier GPU constraints.
- Amharic underperformance: with only 1,845 training examples and a distinct Ge'ez script, Amharic is expected to perform significantly worse than English subsets. Per-language evaluation (Experiment 8) is needed to quantify this gap.
- ROUGE limitations: ROUGE measures lexical overlap and does not capture semantic correctness. A generated answer that is medically accurate but phrased differently from the reference can receive a low ROUGE score. This is especially problematic for health QA, where paraphrase and cultural framing vary widely.
- Dataset noise: at least one mislabeled example was identified during qualitative inspection (syphilis question with herpes reference answer). The scale of label noise in the full dataset is unknown and may affect evaluation reliability.
- Max output length : 128 tokens likely truncated longer answers particularly in Akan. This affects ROUGE scores independently of model quality.

7 Ethical Considerations and Responsible AI

This project operates at the intersection of AI and public health in African communities a domain with significant ethical stakes that warrant careful consideration

7.1 Health misinformation risk

The model generates free form health answers on sensitive topics including HIV, STIs, maternal health and reproductive health. Unlike a retrieval based system that cites verified sources a generative model can produce factually incorrect information with high linguistic fluency a phenomenon known as hallucination. In a health context, a convincing but incorrect answer about medication dosages, STI symptoms, or contraception could cause direct harm to a user. This model is a research prototype and must not be deployed in any clinical or patient-facing context without rigorous medical review and fact-checking by qualified health professionals.

7.2 Low-Resource Language Bias

The dataset is heavily imbalanced toward English (approximately 60% of training data), and within African languages, Amharic is significantly underrepresented (1,845 examples vs 7,624 for English-Uganda). This means the model will produce substantially better answers for English speakers than for Amharic, Akan, Luganda, or Swahili speakers precisely the communities the system is meant to serve most. This is an example of the well-documented

problem of AI systems performing worst for the groups most in need of their benefits. Any deployment of this system should be accompanied by per-language quality audits and clear communication to users about language-specific limitations.

7.3 Cultural and Contextual Appropriateness

Health communication norms vary significantly across cultural contexts. A response appropriate for one community (e.g., direct language about sexual health in an urban English-speaking context) may be inappropriate or ineffective in another. The LLM-as-a-Judge metric used in this competition includes language appropriateness as a criterion, but this automated metric cannot fully capture cultural nuance. Human evaluation by community health workers familiar with each target language and culture would be essential before any real-world deployment.

7.4 Responsible Use Statement

This model is developed solely for academic purposes as part of a machine learning course. It should not be used to provide health advice to real individuals. Any future development building on this work should involve collaboration with public health experts, community health workers, and native speakers of the target languages from the earliest stages of design.

8. Lessons Learned

- fp16 with mT5 is unreliable on T4 GPUs: always test with fp32 first, then try bf16 if available (A100/H100 GPUs support bf16 more stably than T4s support fp16).
- Save checkpoints sparingly: mT5-small checkpoints are ~1.2GB each. With `save_total_limit=2` and `save_steps=500`, two failed experiments consumed the entire 20GB Kaggle working directory quota. Using `save_strategy = 'no'` and saving only the final model avoids this.
- multi-GPU DataParallel can cause OOM with fp32 : running batch size 8 across 2 T4s(effective 16) with fp32 triggered OutOfMemory Error during the backward pass. Reducing to batch size 4 resolved this
- Colab free tier is too unreliable for sustained ML experiments: session disconnections and GPU quota limits make it unsuitable for training runs longer than 1 to 2 hours. Kaggle 's guaranteed 30 GPU hours per week is more suitable for this kind of project.
- EDA first, train second: inspecting language distribution, answer lengths, and data quality issues before training revealed the Akan truncation problem and the Amharic difficulty level, which directly informed experimental design.
- Language-tag prefixes are low-cost and high-value: a simple string concatenation at preprocessing time provides explicit language conditioning signal at zero additional training cost.

9. Future Improvements

- Larger model: fine-tuning mT5-base or mT5-large would likely yield significant quality improvements for all languages, particularly for longer-form Akan answers.
- LoRA/QLoRA: parameter-efficient fine-tuning would enable experimenting with much larger models (e.g., Aya-101, BLOOMZ-1b7) within the same GPU constraints, since only a small fraction of weights are updated.
- Dedicated Amharic handling: pre-training or continued pre-training on Amharic text before fine-tuning on this dataset could significantly improve performance on the most underrepresented language.
- Retrieval-augmented generation (RAG): grounding generated answers in verified health knowledge bases (e.g., WHO fact sheets translated into target languages) would reduce hallucination risk.
- Human evaluation: ROUGE-based metrics are insufficient for health QA. Future work should include evaluation by native-language health professionals to assess factual accuracy, cultural appropriateness, and clinical safety.

10. AI Usage Disclosure

In accordance with the academic integrity requirements of this course, the following describes where and how AI assistance was used in this project:

- Claude (Anthropic) was used as a debugging and tutoring assistant throughout the project: it helped diagnose environment-specific errors (fp16 NaN collapse, DataParallel OOM, disk quota overflow), suggested fixes for HuggingFace API changes (processing_class= replacing tokenizer= in Seq2SeqTrainer), and explained the underlying causes of each issue.
- Claude also helped draft boilerplate training code, which was then adapted, run, and verified by the student. All experimental decisions, hyperparameter choices, and result interpretations are the student's own.
- This report was drafted with Claude's assistance as a structural tool, but all analysis, findings, and critical discussion reflect the student's genuine understanding of the implementation and results as demonstrated by the ability to explain every decision in the demo video.
- No AI tool was used to fabricate experimental results, generate synthetic leaderboard scores, or produce content that was not based on actual runs performed by the student.

References

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Appendix: Project Deliverables

1. Project Repository:

<https://github.com/utatsineza/Multilingual-health-QA-fine-tuning-project>

2. Demo

Video:https://drive.google.com/drive/u/1/folders/1IsUNC-Ya6SIBfJeV-WfBGhYVsTS_mbiO?t=hs=true